

Data Privacy and Security in Health Technologies: Challenges, Risks, and Regulatory Implications

Abstract

Digital health tech, like electronic health records, wearables, apps, telemedicine and AI, has really changed how healthcare is given. Now, we can watch patients in real time, create plans that fit each person, and make things run smoother behind the scenes. But, these new tools also make a ton of personal health info, things like biometric data, DNA, habits, and where people are, and that can cause issues. While these improvements could lead to better health and a more efficient system, they also open the door for privacy problems. Things like data leaks, sharing info without permission, biased algorithms, and holes in the rules become larger concerns. (World Health Organisation, 2023).

This review of articles looks at research papers, rules, and reports from 2016 to 2026. These were found on PubMed, Scopus, Google Scholar, and health organization websites. One thing we found is that online security threats are getting worse. In 2024, there were 725 big data breaches in U.S. healthcare alone, which put 277 million patient files at risk. That's up 65% from 2023. Current rules, like HIPAA and GDPR, don't work well with today's apps, data moving across countries, and AI that's hard to understand. The review showed problems like too much data being collected, not enough patient approval, and vendors not being responsible enough.

To fix this, we suggest designing with privacy in mind, having required check-ups by outside companies, teaching patients more, and making international rules that are the same everywhere. To make this work, people in charge, healthcare workers, tech companies, and patients need to work together. This way, new tech in healthcare will respect patient privacy and still provide the benefits it promises. (Price, 2019).

Introduction

1.1 The Digital Health Revolution

Digital tech has totally changed how healthcare works. Now, doctors can keep all your records in one place online with Electronic Health Records. You can even talk to a doctor from home using telemedicine. Wearable gadgets keep track of your health stats all day long, and there are phone apps to help you take care of yourself. Plus, computers are getting smarter and can help doctors figure out what's wrong faster. Guess what? The health app market made a whopping \$300 billion a year by 2025, and people downloaded health apps over 7 billion times. The Apple Watch and Fitbit tracked 1.5 billion hours of health data in 2024 alone (BGO Software, n.d.).

These technologies use huge collections of data, including medical records, lab results, gene sequences, location data, sleep info, and behavior patterns. Analyzing data as it comes in allows for predictions, early action, and custom medicine. Remote monitoring of patients cut hospital readmissions by 38% in Medicare tests. Also, AI radiology tools spotted breast cancer with 94% accuracy, doing better than human radiologists (Medintelx, 2024).

1.2 The Privacy Paradox

This huge amount of data is creating big privacy problems. Health info is worth a lot on the dark web, a full medical record can get \$250, while stolen credit cards only get about \$5. In the U.S., there have been 6,759 health care data breaches since 2009, exposing 847 million patient files. 2024 was the worst year yet, with 725 breaches that risked 277 million files. One really terrible example is the Change Healthcare ransomware attack, which messed up prescription services for a third of people in the U.S. (Sprinto, 2025).

After a data breach, patients quickly lose trust. A 2024 study showed that 67% of patients would hide info from digital health devices because they're worried about their data being exposed, which hurts how well these tools work. Consent forms are a mess, privacy policies are super long, like over 10,000 words, but written simply, so people just agree without understanding. It's even harder to get real consent with AI because patients can't understand how these computer programs make choices (Antaranews, 2025).

1.3 Research Objectives

This paper systematically investigates:

- 1. Primary privacy and security challenges across health technologies
- 2. Effectiveness and limitations of current regulatory frameworks
- 3. Practical solutions balancing innovation with patient rights

1.4 Paper Structure

Following this introduction, Section 2 reviews relevant literature, Section 3 details methodology, Section 4 presents findings, Section 5 offers discussion and recommendations, and Section 6 concludes.

Table 1: Digital Health Technologies and Privacy Risks

Technology	Data Collected	Primary Privacy Risks
EHRs	Medical history, labs, medications	Ransomware, unauthorized access
Wearables	Heart rate, sleep, GPS	Data broker monetization
mHealth Apps	Habits, biometrics	Residual device data
AI Diagnostics	Genomics, imaging	Re-identification, algorithmic bias

Literature Review

2.1 Evolution of Privacy Concerns

As tech has gotten better, worries about keeping health data private have changed as you might expect. Before 2000, when everything was on paper, people worried about someone just walking in and seeing things they shouldn't. When electronic health records became common around 2009, thanks to the HITECH Act, new problems came up with computer security. Then, when smartphones took off in the 2010s, it became easier than ever to track people's health info all the time. Now, with AI, we need huge amounts of data, and that means there's a greater chance someone could figure out who a person is from that data, even if it's not obvious. (Theodos, 2021).

2.2 Cybersecurity Threats

Healthcare is a major cybersecurity target. Health records are worth 50 times more than financial info on the dark web. This is because they can be used for things like insurance scams, identity theft, and blackmail throughout a person's life. When we look at data breaches, we see a steady increase in problems.

Table 2: U.S. Healthcare Breach Trends (2020-2025)

Year	Breaches	Records Exposed	Avg Records/Breach
2023	541	168M	310K

Year	Breaches	Records Exposed	Avg Records/Breach
2024	725	277M	382K
2025*	312	145M	465K

Through September 2025 (Sprinto, 2025);

In 2024, almost 80% of data breaches were due to hacking or IT problems. Ransomware attacks really took off, hitting almost two a day by 2023. The Change Healthcare hack in February 2024 was huge, the biggest healthcare breach ever. It exposed data on 190 million Americans, messed up pharmacy payments across the country, and cost UnitedHealth \$872 million in immediate costs, plus another \$2.45 billion in lost income (HIPAA Journal, 2025).

2.3 Consent and Data Collection Practices

Health apps often grab extra info that isn't really needed for your health. For example, fitness trackers keep tabs on where you are, along with your heart rate, supposedly to map your activity. Mental health apps might snag your device ID, contacts, and what you've been browsing. Privacy policies are super long, like a novel, and full of confusing legal stuff. Most people (91%) never even bother reading them, and over half just agree without looking. (Sharon, 2016).

Even after you uninstall an app, some data sticks around. Mobile apps leave behind traces that can be found by looking at your device, like stored biometric info and how you use the app. Also, services like 23andMe can figure out who your relatives are, even if they didn't agree to it, by matching your data with others(Theodos, 2021).

2.4 AI and Algorithmic Privacy Risks

AI makes privacy problems worse in a few main ways: it can figure out who people are even when data is supposed to be anonymous, it can spread biases, and it can be hard to understand how it works. For example, supposedly anonymous datasets can be re-identified with almost perfect accuracy by matching them up with public info. Also, if the data used to train AI isn't diverse enough, it can lead to skewed results. Studies have shown that Black patients are more likely to get a wrong negative result in cancer diagnoses, and women are more likely to be misdiagnosed when cardiovascular algorithms are used. (Antaranews, 2025).

2.5 Regulatory Landscape

HIPAA (U.S.)

The rule protects covered entities like doctors and insurance companies, but it leaves out apps and medical software sellers. It's worth noting that third-party vendors were behind 93 million data leaks in 2023, while providers were only responsible for 35 million (Sprinto, 2025).

GDPR (EU)

Health data is seen as special, so you need to get clear permission to use it. You also have to do data protection impact tests and tell people about any data leaks within 72 hours. If you mess up, fines can be as high as 4% of your total worldwide income. Transferring data to cloud storage in the U.S. is still tricky after the Schrems II ruling. AI systems that are risky need to be approved before they're sold, but most health-related computer programs don't fall into that group(GDPR Local, 2025).

Table 3: Major Regulatory Frameworks

Framework	Scope	Key Strengths	Critical Gaps
HIPAA	Providers/Insurers	Breach notification	Consumer apps excluded
GDPR	All processors	Consent rights, DPIAs	Cross-border complexity
CCPA	California residents	Data sales opt-out	Business size threshold

2.6 Literature Consensus

Scholars universally identify regulatory obsolescence against app ecosystems, vendor accountability gaps, and AI opacity. Privacy-by-design, differential privacy, and federated learning emerge as consensus technical solutions requiring supportive policy frameworks (Dove, 2018).

Methodology

For this study, we did a systematic review of the research out there, focusing on English articles from 2016 to 2026. We looked in places like PubMed, Scopus, and Google Scholar, searching for things like health data privacy or healthcare cybersecurity along with wearable, mHealth, artificial intelligence, or electronic health record. We made sure to dig through official papers from groups like the HHS Office for Civil Rights, the European Data Protection Board, and the World Health Organization. We also checked out industry stuff from HIPAA Journal, Sprinto Healthcare Stats, and HIMSS Analytics.

At first, we found 847 things, but after checking the titles and summaries, we read 152 all the way through. In the end, we used 45 sources that fit what we were after: studies about keeping health tech data safe (22), analyses of rules and policies (14), and stats on data breaches (9).

We sorted what we found into four main areas: how data is gathered, what cyber threats are out there, possible problems with algorithms, and where the rules don't quite cover things. As we went, we kept comparing stuff to make sure we had a good handle on the themes. To make sure our method was solid, we used at least two sources to confirm each point, made sure our stats were fairly recent (from 2023 at the earliest), and didn't include sales pitches from companies that hadn't been checked by other experts.

Findings and Analysis

4.1 Excessive Data Collection

It looks like health apps grab way more info than they need. Fitness apps track your exact location and steps, while sleep trackers figure out where your bedroom is using Wi-Fi. Mental health apps even sneak a peek at your contacts and texts. A study shows that most popular health apps ask for at least three permissions they don't really need, like your location, contacts, or microphone. And to top it off, they use sneaky tactics in their permission requests, like pre-checked boxes and confusing wording, to trick you into saying yes (Sharon, 2016).

Post-installation behavioral tracking persists indefinitely. uninstallation fails to remove cached biometrics, usage analytics, or device fingerprints. Forensic recovery from "deleted" health apps succeeds 87% of time, enabling persistent surveillance profiles (Theodos, 2021).

4.2 Escalating Cybersecurity Threats

Healthcare cybersecurity exhibits catastrophic failure patterns. 2024 established breach records across all metrics:

- **725 large breaches** (500+ records)
- **277 million Americans** exposed (1/3 population)
- **Average 758,000 records/day** stolen
- **\$10.1 million average breach cost** (300% industry average)
- **Ransomware attacks/day: 2.0** (Sprinto, 2025)

The Change Healthcare hack in February 2024 shows how weak systems can be. Hackers got in using stolen Citrix login info, moved around to over 30 systems, and stole 190 million patient records. Then, they used ransomware to encrypt prescription processing all over the country. UnitedHealth paid a \$22 million ransom and lost \$2.45 billion. They're also dealing with over

140 lawsuits. Even after all that, patient data is still being sold on the dark web (HIPAA Journal, 2025).

Working with business partners can be risky. Even though third-party vendors only account for 23% of the notices, they were the reason for 67% of the huge data breaches in 2024. It's easy for people to take advantage of the situation if you have weak HIPAA business associate agreements, don't do enough penetration testing, and are unclear about who is responsible for what.

4.3 AI-Driven Privacy Violations

Machine learning can make privacy problems way bigger because they can do so much and are so complex. For example, it's been shown that you can figure out who people are even when data is supposed to be anonymous almost all the time (like 99.98%!). This is done by matching up the anonymous data with things like voter lists, property info, and what people post on social media. Also, if the data used to train these systems isn't diverse, the systems can be biased. Systems trained mostly on data from white men screw up diagnoses for women way more often (like 34% more). And for Black people, skin cancer tests are much more likely to give a wrong result (20% more) compared to white people (Antaranews, 2025).

AI systems are often like black boxes because of how their code is set up. Deep learning networks use tons of data in ways that are hard to understand. Doctors can't explain most AI treatment advice, about 78% of the time, which can be a problem for getting proper consent from patients. People can't challenge AI decisions if they don't know where the AI's information came from or how it makes choices. This can lead to unfair results.

4.4 Regulatory Enforcement Failures

HIPAA isn't really cutting it in today's digital health world. The way it defines covered entity leaves out a ton of stuff people use, like most apps, at-home genetic tests, and wearable devices you can buy at the store. Plus, the rules for business associates are hard to enforce. Even when these companies break their contracts, the OCR only penalized 12% of them in 2024.

GDPR enforcement is kind of a mess because different countries interpret things differently. Authorities in different member states have made different decisions about the same app breaking the rules. And after the Schrems II ruling, it's still tough to transfer health data between the EU and the U.S., even if companies use standard contract clauses. Also, a lot of healthcare algorithms are considered lower risk by the EU AI Act, even about 68% of them, so it's hard to know if they need examination or not (GDPR Local, 2025).

It's easy to exploit differences in global rules. Places with weak rules are full of health data centers that handle European and American patient info, making it hard to apply laws properly. Because different countries enforce rules differently, international patients don't have one clear way to get help if something goes wrong.

Discussion and Recommendations

5.1 Stakeholder Implications

Patients can be immediately harmed through things like identity theft or job discrimination. The system also suffers because people don't want to use digital health options as much. Healthcare providers get hit with huge costs from breaches, like \$10 million on average, which takes away from patient care. Tech companies find it hard to come up with new stuff because the rules don't always make sense. And those in charge of making the rules have a hard time keeping up with how fast tech is changing.

5.2 Privacy-by-Design Framework

Technical solutions must embed privacy throughout system architecture:

1. **Data Minimization:** Collect only clinically essential data with granular consent
2. **End-to-End Encryption:** Zero-knowledge protocols preventing vendor access
3. **Federated Learning:** Model training without centralized data aggregation
4. **Homomorphic Encryption:** Computations on encrypted health data
5. **Differential Privacy:** Statistical noise preventing re-identification (BGO Software, n.d.)

5.3 Vendor Accountability Regime

Business partners need to do a security test every year, report any problems right away, and have at least \$5 million in cyber insurance. Also, HIPAA should include anyone who handles health data, even if they're not making money from it, in its definition of covered entity.

5.4 Patient Empowerment Initiatives

To help people understand how their data is used, public education should use easy-to-understand charts and graphs. Consent forms need to be approved by regulators to make sure they're written in simple language (at a sixth-grade level or lower) and give people specific control over their data. Also, people should get yearly data receipts that explain what data was gathered, who it was shared with, and how it was used for profit, so they can make smart choices.

5.5 Harmonized Global Standards

The OECD's health data rules can help countries agree on things like data sharing. If countries want to share data across borders, they need to make sure their cybersecurity measures are up to par. Also, having international rules for reporting data breaches stops companies from exploiting loopholes in different countries' regulations.

Table 4: Multi-Stakeholder Solution Matrix

Solution	Patient Benefit	Provider Benefit	Developer Cost	Implementation Timeline
Privacy-by-Design	Granular control	Regulatory compliance	High upfront	12-18 months
Vendor Audits	Breach prevention	Risk transfer	Medium annual	6 months
Consent Reform	Informed choice	Legal protection	Low	3 months
Global Standards	Borderless care	Simplified compliance	Policy only	24 months

Conclusion

Digital health tech could really change how we get care, but privacy issues are a big problem that could stop people from using it. This review shows that cyber attacks are getting worse (with 277 million records exposed just in 2024), consent forms don't work well, it's hard to know how algorithms make decisions, and the rules are out of date when you consider apps and how complex AI is.

Privacy is super important for digital health to last. If we don't make big changes, people won't want to share their info, doctors won't use the tech as much, and companies will be careful about creating new things, meaning we all miss out. There are ways to fix this, like using tech that protects privacy, updating the rules, and making consent easier to understand.

We need to make sure all health apps are designed with privacy in mind, do regular security checks, make consent forms very specific, and create international rules for data. In the future, we should study new tech like blockchain and special types of machine learning that protect privacy.

If we really care about privacy, digital changes in health care can work well. People are sharing personal info like their biometrics, DNA, and habits, trusting that it will be safe. We need to protect that info carefully. Good privacy rules can turn a possible bad situation into a good one, where we get better medicine without giving up our privacy. (Price, 2019).

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